IILM UNIVERSITY

Greater Noida, Uttar Pradesh | School of Computer Science & Engineering

INTERNSHIP PRESENTATION

GenAI-Powered Data Analytics for Predictive Collections Strategy

Predicting customer delinquency and designing a responsible, AI-driven collections strategy for Geldium

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PROGRAM	B.Tech CSE – Specialization in AI & ML with IBM-ICE
SEMESTER	5th Semester
ORGANIZATION	Tata iQ — GenAI Powered Data Analytics Job Simulation (via Forage)

OVERVIEW

Agenda

ABOUT THE ORGANIZATION

Tata, Tata iQ & Forage

TATA

One of India's largest and most diversified business conglomerates, with operations spanning information technology, financial services, steel, automotive, and retail across more than 100 countries.

TATA iQ

The analytics and artificial-intelligence arm of the Tata ecosystem, applying data science, machine learning, and Generative AI to solve real business problems — including financial-services risk and collections management.

FORAGE

A global virtual work-experience platform that delivers Tata's job simulation — free, self-paced, employer-designed tasks that mirror real analyst work.

THE SIMULATION

GenAI Powered Data Analytics Job Simulation

- Completed: August 2026
- Simulated client: Geldium (consumer lending company)
- Role: Data Analyst supporting the Collections function
- 4 practical, business-facing tasks
- Certificate of Completion issued by Forage on behalf of Tata

THE BRIEF

Problem Statement

Geldium's collections team lacked a systematic, data-driven way to identify at-risk customers early and prioritize outreach — relying instead on reactive, post-due-date collections.

1

Costly & Reactive

Late-stage collections is expensive to run and damages the customer relationship.

2

Signals Missed Early

A purely rules-based approach misses combined behavioural + financial risk signals.

3

The Ask

Build a predictive model to flag delinquency risk early, and translate it into a responsible, business-ready collections strategy.

GOALS

Project Objectives

1. Assess data quality and resolve missing values before modelling
2. Surface early behavioural and financial risk indicators via EDA
3. Design a predictive pipeline: feature selection, model choice, evaluation
4. Translate model insights into a SMART, business-ready recommendation
5. Design an agentic, AI-driven collections strategy with human oversight
6. Apply fairness, explainability & regulatory-compliance principles throughout

Dataset Overview

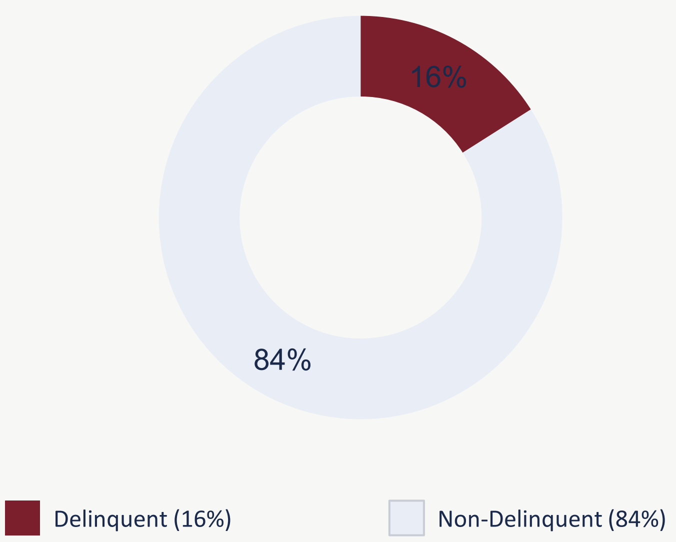
500
Customer records

19
Fields per record

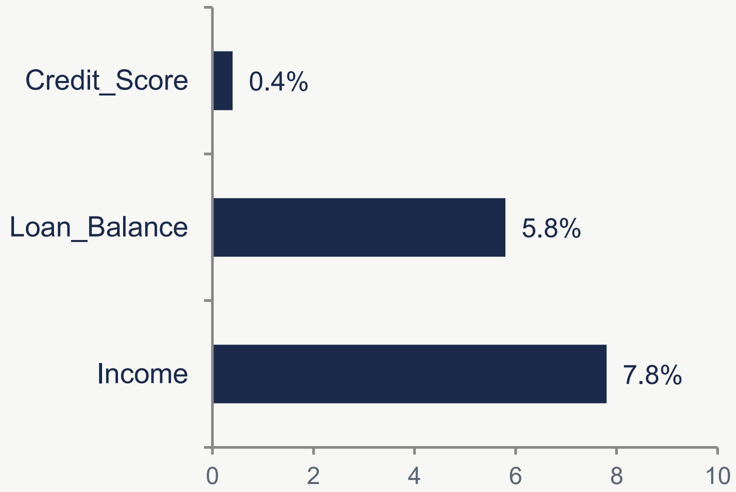
16.0%
Base delinquency rate

0
Duplicate IDs found

Delinquency Split



Missing Values by Field



KEY QUALITY NOTES

- Missing values resolved via grouped median imputation
- Employment_Status labels standardized (6 → 4 categories)
- 4 records show >100% credit utilization — flagged for confirmation

Key Risk Indicators

Employment Status	Unemployed customers show the highest delinquency (19.4%); retired customers the lowest (11.5%).
Credit Card Type	'Business' cardholders show the highest delinquency (21.3%); 'Platinum' the lowest (11.8%).
Location	Delinquency ranges from 12.0% (New York) to 19.6% (Los Angeles) across five cities.

THE UNEXPECTED FINDING

Weak univariate correlation

None of the core numerical variables — income, credit score, utilization, missed payments — show meaningful linear correlation with delinquency (all coefficients near zero).

Working hypotheses:

- Risk may be driven by non-linear interactions (e.g., high utilization × low income)
- The simulation's target may be generated independently of listed features

TASK 2 — PREDICTIVE MODELLING

Modelling Approach

A two-stage approach: logistic regression as an interpretable baseline, benchmarked against gradient-boosted trees to capture non-linear interactions.

PRIMARY

Logistic Regression

- High interpretability — transparent coefficients
- Well-established in credit scoring & regulation
- Lightweight, fast to retrain and monitor

CHALLENGER

Gradient-Boosted Trees

- Captures non-linear feature interactions
- Typically higher accuracy on complex risk data
- Needs SHAP/feature-importance for explainability

TOP 5 PREDICTIVE FEATURES

Missed Payments (12mo)	**Credit Utilization**	**Debt-to-Income Ratio**	**6-Month Payment Trend**	**Credit Score**

TASK 2 — PREDICTIVE MODELLING

Evaluation Strategy

With only 16% of customers delinquent, accuracy alone is misleading — a model that always predicts "not delinquent" would score 84% while catching zero at-risk customers.

1

Precision

Limits wasted outreach to good customers

2

Recall

Priority metric — missing at-risk customers costs more than a false alarm

3

F1 Score

Balances precision and recall in one score

4

AUC-ROC

Threshold-independent view of separability

FAIRNESS & BIAS CHECKS

Segmented performance review across Employment_Status, Location & age bands • Proxy-variable review • Disparate-impact ratio (80% rule) • Reweighting & threshold adjustment if disparities are found

The SMART Recommendation

Launch a proactive outreach program that flags customers crossing a 70% credit-utilization threshold — before a payment is ever missed.

S

Specific

Target customers whose utilization exceeds 70% for two consecutive billing cycles

M

Measurable

Track flagged-group delinquency vs. a comparable unflagged control group each quarter

A

Actionable

Route flagged accounts to the early-engagement queue via existing channels

R

Relevant

Directly addresses the model's top predictor; shifts collections from reactive to preventive

T

Time-bound

One-quarter pilot; expand if delinquency drops ≥10% vs. control

Ethical Guardrails

F

Fairness

Exclude protected attributes; monitor outcomes by demographic group; rely on behavioural variables over demographic proxies.

E

Explainability

Communicate predictions with plain-language drivers, not raw model scores, so the Collections team can act on the reasoning.

R

Responsible Use

The model triggers supportive outreach and counseling, not automatic penalties — a false positive means a helpful check-in.

T

Transparency

Model performance and flagged segments are reviewed regularly; customer data is used only for the stated purpose.

These four safeguards run continuously alongside the model — not as a one-time check before launch.

Agentic System Architecture

A closed feedback loop: every action the system takes feeds back into the model, so it gets smarter and fairer over time.

1 Customer Data

Repayment history, utilization, engagement, account status

→

2 Decision Logic

Risk-scoring model + business-rule overlay assigns a risk tier

→

3 Automated Action

Tiered outreach — reminder, offer, or human referral

→

4 Learning Loop

Outcomes tracked and fed back to retrain & recalibrate

AUTONOMOUS

Reminders • nightly re-scoring • routing accounts • pre-approved payment offers • logging outcomes

HUMAN-IN-THE-LOOP

Hardship & forbearance approval • settlement negotiation • dispute review • credit-reporting or legal actions

TASK 4 — BUSINESS CASE

Expected Business Impact

↓

Delinquency

Fewer accounts rolling to 30/60/90+ days past due

↑

Repayment Rate

More on-time and self-cured resolutions

↓

Cost-to-Collect

Automation absorbs routine outreach volume

↑

Scalability

Handle portfolio growth without adding headcount

CUSTOMER OUTCOMES

- More empathetic, right-sized outreach for each customer's situation
- Faster, clearer answers thanks to explainable decisions
- More consistent, equitable treatment across customer segments
- Greater trust in Geldium as a fair and transparent lender

REFLECTION

Skills Gained

1. Data-quality assessment, cleaning & justified imputation strategies
2. Exploratory data analysis and behavioural risk-factor identification
3. Predictive-model planning: feature selection, model choice, evaluation
4. Business storytelling using the SMART recommendation framework
5. Designing agentic AI workflows with human-in-the-loop governance
6. Responsible & transparent use of Generative AI in a professional workflow

KEY TAKEAWAY

Scale insight into action without losing the human touch.

An agentic AI collections system lets Geldium act on risk signals at scale, while fairness, explainability, and human oversight keep every decision defensible.

Thank You

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